

Analysis of
Nanopore MinION sequencing data
with
EPI2ME
Bacterial assembly and annotation workflow
on Windows computer

Version: April 2025



Minimal Requirements for Computer/Laptop to run Epi2ME analysis workflow:

Compute requirements

Recommended requirements:

- CPUs = 16
- Memory = 64GB

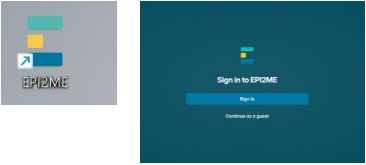
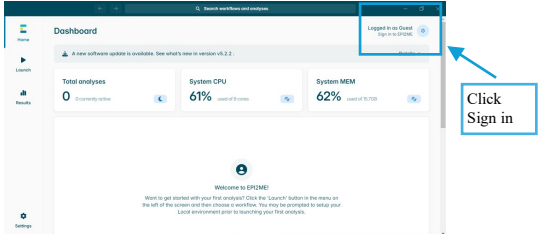
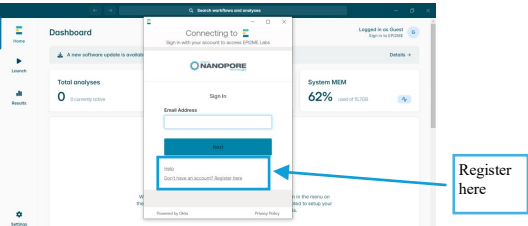
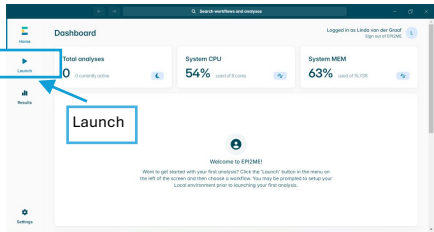
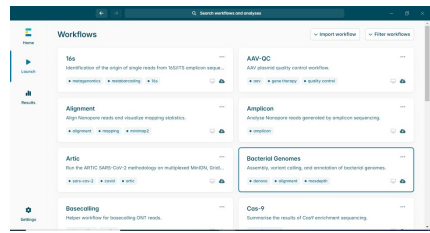
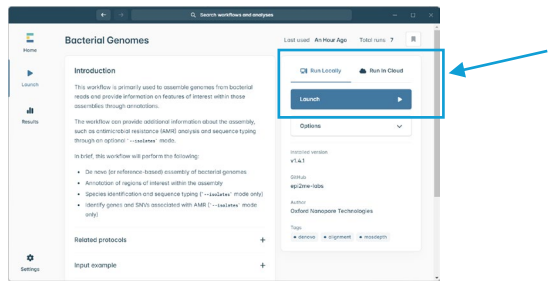
Minimum requirements:

- CPUs = 8
- Memory = 32GB

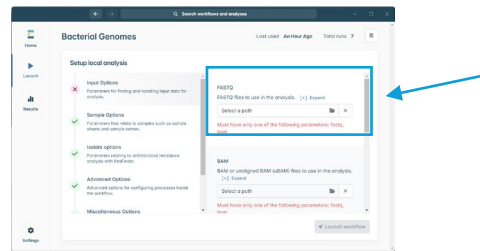
Approximate run time: 20-40 minutes per sample with ~50x coverage using minimum requirements

ARM processor support: True

Manual for analysis of Nanopore MinION sequencing data with Epi2ME software on Windows computer

Open EPI2ME Software and Sign in If you do not have an account, click “Continue as a guest”	
As guest: Click “Sign in to EPI2ME”	
As guest: Click “Don’t have an account? Register here” Register and login	
After login, click “Launch”	
Click on “Bacterial Genomes”	
If the software is correct installed, the “Launch” button will be dark blue. If the software is not correct installed, the red button “Setup required” will be visual. In this case, please install EPI2ME software as described in Manual “Install EPI2ME Software”	
Click on “Launch” button	

At FASTQ section, click on
“Select a path”

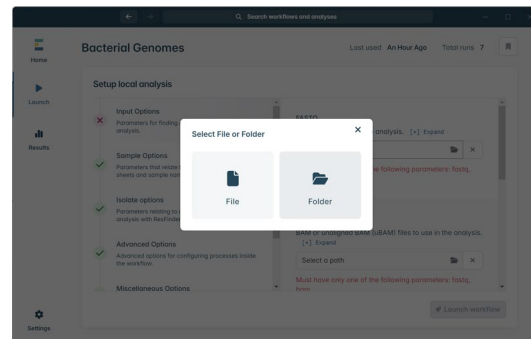


Click on Folder and navigate to folder
with Nanopore Sequence Data

Select Folder “fastq_passed”

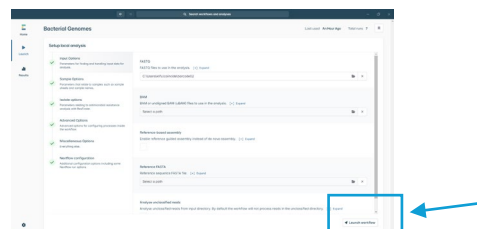
Note 1: the complete folder with all
barcodes can be processed in 1 run. This
will take several hours.

Note 2: We had an error once, and the
whole run was stopped due to a error
(number 1) of Flye of an empty
barcode. The solution was to delete the
empty barcode folders in de
“fastq_passed” folders.

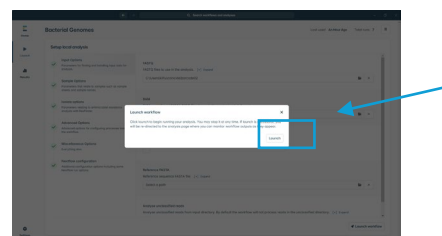


Do not select BAM, reference-based
assembly and Reference fasta files

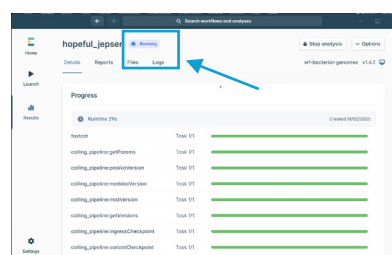
Click “Launch Workflow”



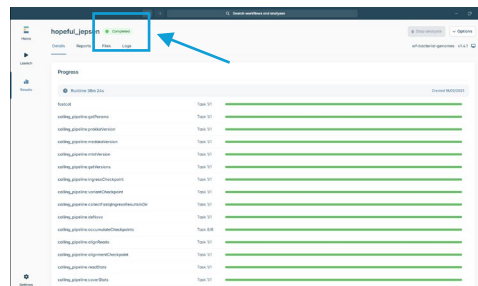
Click “Launch” and workflow will start



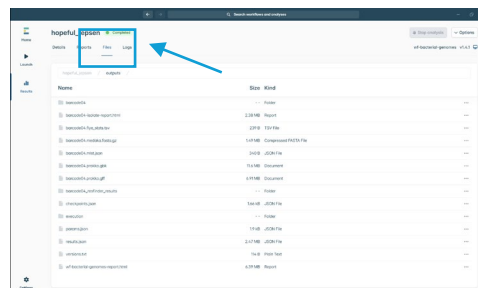
Progress screen with running analysis



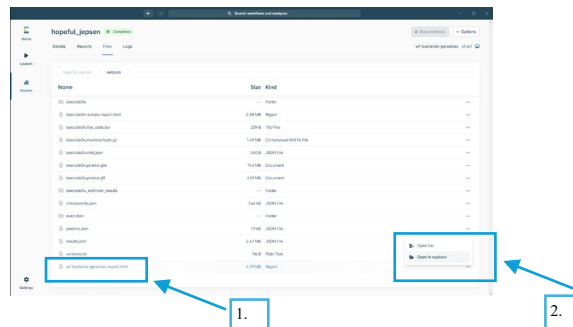
Example of completed analysis



Navigate to “Files”

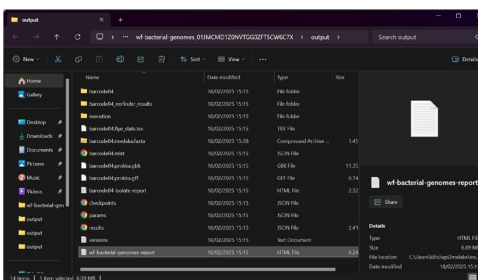


1. Go to file
“wf_bacterial_genomes_report.html”
2. Open file in Explorer



In Explorer:

1. right mouse click on wf-bacterial-genomes-report.html file
2. select “open with”
3. select browser, for example google chrome

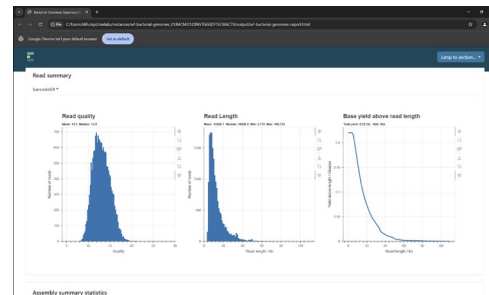


Bacterial Genomes Analysis Summary report is shown in browser



Asses the read summary.

Read length should be at least 2kb

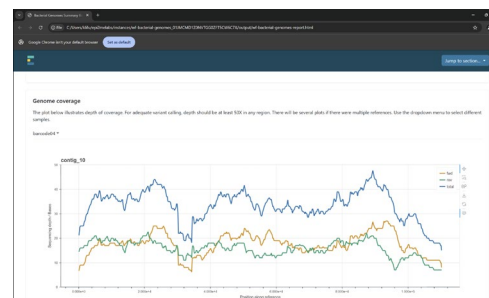


Assembly summary statistics

Multiplicity is how many times a particular edge appears in the genome (it is calculated based on coverage)

Contig	Length (bp)	Coverage	Circular	Repeat	Multiplicity
contig_1	489019	35	Y	N	1
contig_2	240707	33	Y	N	1
contig_3	112280	36	Y	Y	3
contig_10	110000	37	Y	N	1
contig_5	48870	139	Y	Y	4
contig_6	9000	140	Y	Y	4

Coverage per contig can be assessed



Annotations

Gff and gbk files generated by Prokka also available in files in “outputs” folder

Contig	Start	End	Strand	Gene	Product	NC number
contig_8	174	1010	+	gpi_1	P1 protein	-
contig_8	1030	1406	+	-	Hypothetical protein	-
contig_8	1933	3986	+	-	Hypothetical protein	-
contig_8	2066	2269	-	-	6S body transcription (S6S)	-
contig_8	2316	2338	+	-	Hypothetical protein	-
contig_8	2325	2335	-	-	Hypothetical protein	-
contig_8	2607	4227	-	-	Hypothetical protein	-
contig_8	4033	1213	+	hsl_2	Protein translocator (Hsl) DNA-membrane bond	-
contig_8	5388	5409	+	-	Hypothetical protein	-
contig_8	1000	6410	-	hsl_1	Heat shock protein 70	33224

Resfinder results are also available as text file in “outputs” folder (*_resfinder_results)

NCBI Gene database entry for the gene **schw** (Schwann cell marker 4). The page displays the gene's location on chromosome 17, its sequence, and a table of variants. The variant table has columns for Name, Position, and Version, and lists a single variant named 'pindel' at position 1,143.

Name	Position	Version
pindel	1,143	

Google Chrome can't find the default browser. [Learn more](#)

Google Chrome can't find the default browser. [Learn more](#)

[Jump to software](#)

Software versions

12 - 1 entries per page

Search...

Name	Version
glibc	1.5.1
musl	2.0.9
glibc	2.26
glibc	2.26
libc	0.16
musl	0.7
glibc	2.9.5-10001
glibc	2.15

Showing 1 to 8 of 8 entries

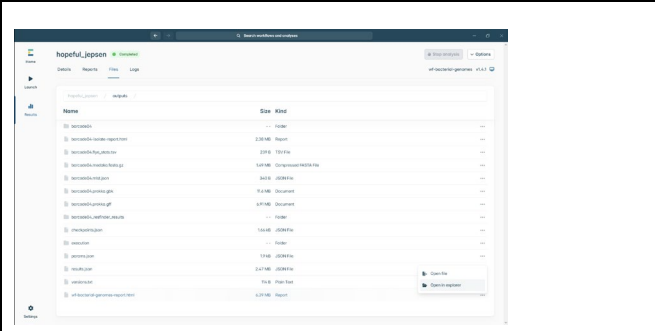
Workflow parameters

12 - 1 entries per page

Search...

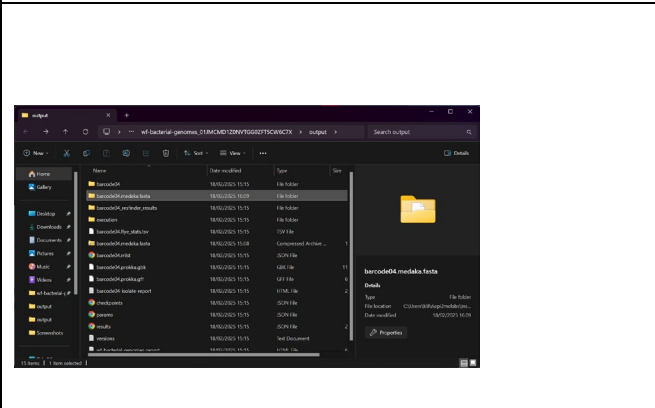
[illegible][illegible]

In “outputs” folder:
*.medaka.fasta.gz is assembled fasta
file

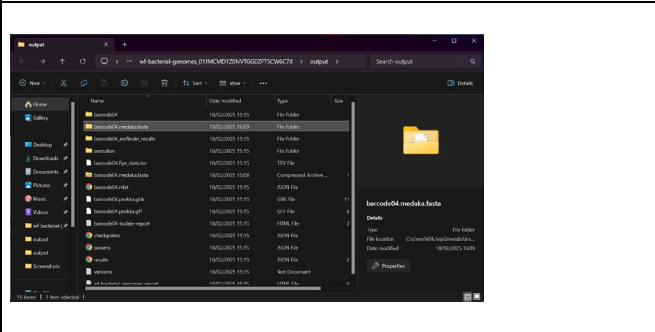


In explorer:

- right mouse click on
barcode*.medaka.fasta.gz
- select Extract All



Extract generates new folder with assembled *.fasta file



ADVANCED EPI2ME installation options – change Workflow files

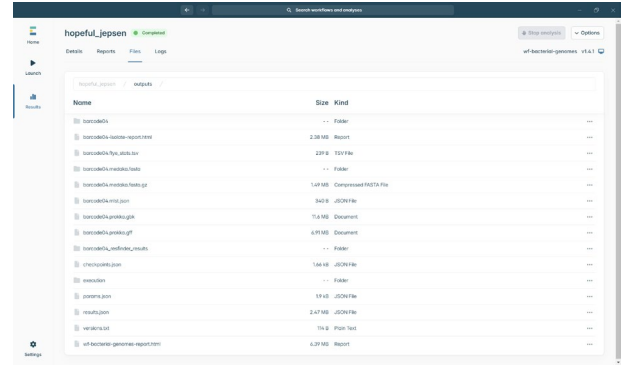
Please note: Described modifications can only be performed by advanced users who know what they are doing

1. Advanced installation; change workflow to “isolates mode”. Default setting is that isolates mode of bacterial genomes workflow is turned off. This isolates mode will add analysis like species identification, MLST STs and Resfinder result.

Follow the following steps to turn isolates mode on.

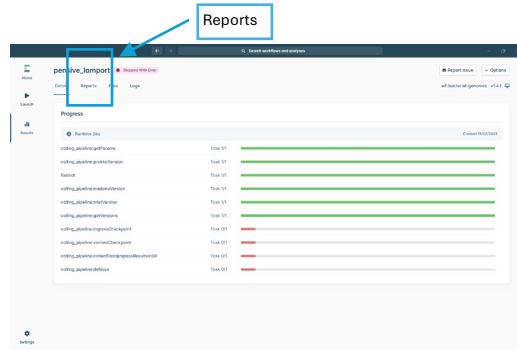
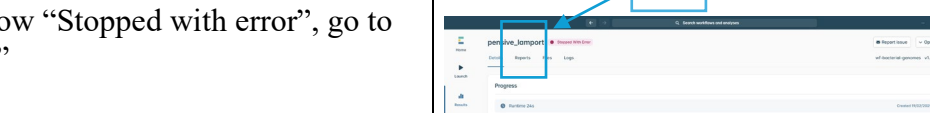
[illegible]

Launch Bacterial Genomes workflow again and check if mlst and resfinder files are generated



2. *Troubleshooting; Workflow stops with notification that not enough memory is available for workflow, while computer/laptop has the prescribed minimal 32gb RAM*

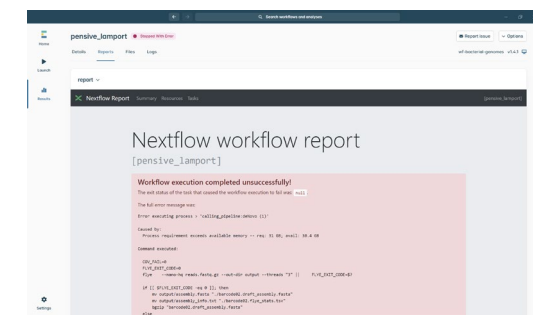
If workflow “Stopped with error”, go to “Reports”



Check error.
If de novo assembly has stopped and
“Caused by: Process requirement exceeds
available memory”...

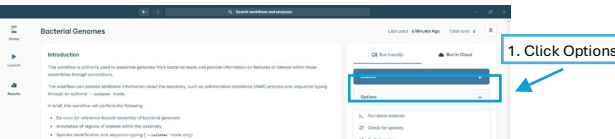
...continue with following steps

If other error is reported, do not continue
with following steps



Navigate back to Bacterial Genomes Workflow

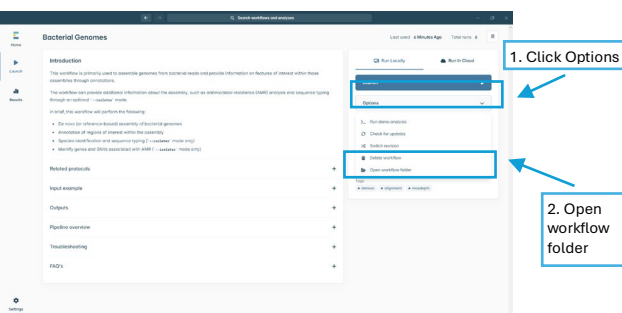
3. Click “Options”
4. Click “Open workflow folder”



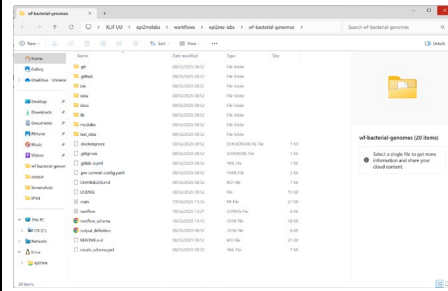
1. Click Options

2. Open workflow folder

3. Click “Options”
4. Click “Open workflow folder”



Open file “main” in text editor



Scroll down to “process deNovo” and change memory “31 GB” to “24 GB”

Save file

[illegible]

Open file “nextflow” in text editor

Scroll down to awsbatch process and change
“memory = 31 G” to “memory = 24 G”

Save file

```
conda (
  conda.enabled = true

  asubatch {
    process {
      executor = 'subbatch'
      queue = "${params.asu_queue}"
      memory = '16G'
      withlabel: of common {
        container = "${params.asu_image_prefix}-of-common:${params.of_common_sha}"
      }
      withlabel: of bacterial genomes {
        container = "${params.asu_image_prefix}-of-bacterial-genomes:${params.of_bacterial_genomes_sha}"
      }
      withlabel: prokka {
        container = "${params.asu_image_prefix}-prokka:${params.of_container_sha_prokka}-root"
      }
      withlabel: medaka {
        container = "${params.asu_image_prefix}-medaka:${params.of_container_sha_medaka}"
      }
      withlabel: raxml {
        container = "${params.asu_image_prefix}-raxml:${params.of_container_sha_raxml}"
      }
      withlabel: minis {
        container = "${params.asu_image_prefix}-minis:${params.of_container_sha_minis}"
      }
      withlabel: sequencer {
        container = "${params.asu_image_prefix}-sequencer2:${params.of_container_sha_sequencer}"
      }
    }
    shell = ['bin/bash', '-mo', 'pipefail']
  }
}
```

Launch Bacterial Genomes workflow again and check if it runs without errors till “Completed”

